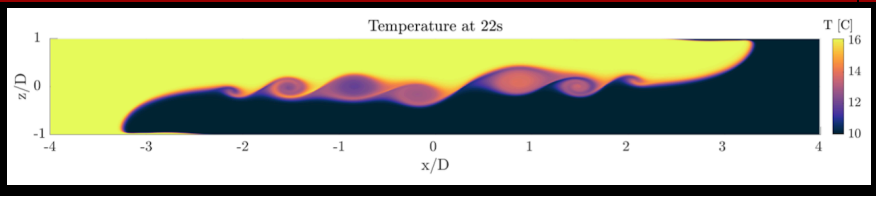
Computational Sciences Research Center

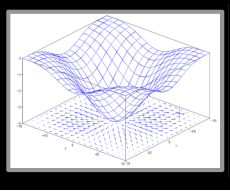
SDSU San Diego State University

MOLE LIBRARY TUTORIAL

May 12, 2026, from 10:00 AM to 12:00 PM (U.S. Pacific Time)
Bioscience Center Auditorium, San Diego State University,
and also accessible online

MOLE





The Computational Science Research Center (CSRC) at San Diego State University invites you to:

A Tutorial on The Mimetic Operators Library Enhanced (MOLE)

May 12, 2026, from 10 AM to 12:00 PM. U.S. Pacific Daylight Saving Time.

The Mimetics Operator Library Enhanced (MOLE) is a high-quality (C++ & MATLAB/Octave) library that implements high-order mimetic difference operators to numerically solve partial differential equations. It provides discrete analogs of the most common vector calculus operators: Gradient, Divergence, Laplacian, and Curl. These operators (highly sparse matrices) act on staggered grids (uniform, non-uniform, curvilinear) and preserve locally and globally quantities of interest for systems of conservation laws.

This tutorial will cover an introduction to MOLE mimetic difference operators; gradient, divergence, Laplacian and curl for C++ and MATLAB/Octave users. Attendants will learn how to use the MOLE library through examples, and also review the installation and guidelines to contribute to the MOLE library. Please, read the ***Getting Started*** Section in mole-docs.readthedocs.io/ and install or clone the MOLE library. Installation support will be provided 90-minutes prior to the beginning of the tutorial.

The tutorial is free and will be held in person at **SDSU's *Bioscience Center Auditorium* on May 12, 2026 from 10 AM to 12:00 PM U.S. PDT, and also online via Zoom**. To register for the meeting in person or online, use the QR above.

For questions about this tutorial contact: jcastillo@sdsu.edu
to subscribe to our mailing list, please send an email to mole-join@engr.sdsu.edu